Possible Directions Around Schiffer's Conjecture, Random Analytic Domains, and Eigenvalue Limits

May 7, 2026

Basic setup and current status

Schiffer's conjecture asks whether a bounded regular domain $\Omega \subset \mathbb{R}^n$ must be a ball if there exists a nontrivial solution of the overdetermined Neumann eigenvalue problem

$$\Delta u + \lambda u = 0 \quad \text{in } \Omega, \qquad \partial_\nu u = 0 \quad \text{on } \partial\Omega, \qquad u|_{\partial\Omega} = \text{constant},$$

for some $\lambda > 0$. In the plane, the conjecture remains open even for general convex domains. Existing results prove the conjecture for special classes: low-index eigenvalues, centrally symmetric convex domains under additional hypotheses, certain analytic geometries, domains close to balls, and some domains for which the associated Pompeiu property can be verified.

The conjecture is tightly related to the Pompeiu problem. A useful equivalent formulation is that a domain fails the Pompeiu property if and only if there is a radius $k > 0$ such that the Fourier transform of its indicator vanishes on the circle of radius k:

$$\widehat{\mathbf{1}_\Omega}(k\omega) = 0 \qquad \text{for every } \omega \in S^1.$$

Thus Schiffer's conjecture can be attacked by studying the nonlinear map

$$\Omega \longmapsto \left(\omega \mapsto \widehat{\mathbf{1}_\Omega}(k\omega)\right),$$

and asking whether the only convex domain for which this function vanishes identically for some $k > 0$ is the disc.

For random Fourier parametrizations of the boundary, the "nonanalytic boundary" shortcut is probably not the right direction if the Fourier coefficients decay exponentially. Indeed, for

$$\Phi(t) = \sum_{n \in \mathbb{Z}} a_n X_n e^{2\pi i n t},$$

if $|a_n| \lesssim e^{-\rho|n|}$ and the random variables X_n have at most subexponential almost-sure growth, then Φ extends holomorphically to an annulus. The boundary is then real analytic almost surely. This places the model directly inside the hard analytic regime of Schiffer/Pompeiu, rather than outside it.

Direction 1: random analytic convex domains as a genericity problem

Instead of using boundary nonanalyticity, one can try to prove an almost-sure Pompeiu property for random analytic convex domains. A natural model is to randomize a support function

$$h(\theta) = h_0(\theta) + \sum_{n \neq 0} a_n X_n e^{in\theta},$$

with the convexity condition

$$h + h'' > 0.$$

This is cleaner than randomizing an arbitrary boundary parametrization, because convexity is encoded by a scalar positivity constraint. One may then define Ω_h as the convex body with support function h.

The target statement would be

$$\mathbb{P}\left(\exists k > 0 : \ \widehat{\mathbf{1}_{\Omega_h}}(k\omega) \equiv 0 \text{ on } S^1\right) = 0.$$

Heuristically, for each fixed k, the condition $\widehat{\mathbf{1}_{\Omega_h}}(k\omega) \equiv 0$ is infinitely many scalar equations in the random Fourier coefficients. One would like to turn this into a rigorous zero-probability statement. The hard point is the existential quantifier over $k > 0$, as well as the fact that the equations are nonlinear and analytic in the boundary.

A possible route is:

1. Work first with finite-dimensional truncations

   $$h_N(\theta) = h_0(\theta) + \sum_{1 \le |n| \le N} a_n X_n e^{in\theta}.$$

   For a fixed k and finitely many test directions ω_j, show that the joint vanishing of $\widehat{\mathbf{1}_{\Omega_{h_N}}}(k\omega_j)$ has probability zero unless the corresponding analytic functions vanish identically.

2. Prove that the relevant analytic functions are not identically zero by differentiating with respect to a carefully chosen Fourier mode.

3. Replace finitely many directions by enough directions to force vanishing on S^1, using analyticity in ω.

4. Control the parameter k by restricting first to compact intervals and then using asymptotic decay/oscillation of $\widehat{\mathbf{1}_\Omega}(\xi)$ as $|\xi| \to \infty$.

This would not prove Schiffer for all convex domains, but it would give an almost-sure Schiffer theorem in a genuinely analytic random class.

Direction 2: compactness of hypothetical convex counterexamples

Another approach is to assume that convex non-disc counterexamples exist and study their limits. Normalize area, barycenter, and perhaps diameter. By Blaschke selection, a sequence of convex bodies has a Hausdorff-convergent subsequence. Suppose Ω_j are convex counterexamples with associated solutions u_j and eigenvalues λ_j:

$$\Delta u_j + \lambda_j u_j = 0, \qquad \partial_{\nu_j} u_j = 0, \qquad u_j|_{\partial\Omega_j} = \text{constant}.$$

If the limit Ω_∞ is nondegenerate and sufficiently regular, then spectral compactness and elliptic estimates might allow the overdetermined condition to pass to the limit. If the limiting domain is not a disc, this gives a limiting counterexample. One could then try to reduce the problem to "minimal" or extremal counterexamples.

The important obstruction is that a sequence of counterexamples could converge to a disc. Then compactness alone gives no contradiction, because discs are known Schiffer domains. Therefore this program must be paired with a local rigidity theorem near the disc:

$$\Omega \text{ sufficiently close to a disc and Schiffer} \quad \Longrightarrow \quad \Omega \text{ is a disc.}$$

There is existing perturbative work in this direction. The useful research question is whether the known local rigidity estimates are strong enough to exclude all normalized convex counterexample sequences converging to a disc. If so, one could obtain a global compactness contradiction under appropriate regularity assumptions.

Direction 3: bifurcation from the disc

The most direct counterexample search is local bifurcation. Discs satisfy the overdetermined problem at radial Neumann eigenvalues. Parametrize nearby convex analytic domains by a radial function

$$r(\theta) = 1 + \varepsilon f(\theta),$$

or, better for convexity, by a support function

$$h(\theta) = 1 + \varepsilon g(\theta), \qquad h + h'' > 0.$$

Then study the nonlinear equation

$$F(h,k)(\omega) := \widehat{\mathbf{1}_{\Omega_h}}(k\omega) = 0 \qquad \text{for all } \omega \in S^1.$$

At $h \equiv 1$ and suitable values of k, this equation is satisfied by the disc. The bifurcation question is whether there are nonconstant branches h_ε solving $F(h_\varepsilon, k_\varepsilon) = 0$.

This converts Schiffer near the disc into a Lyapunov-Schmidt/shape-derivative problem. If the linearized operator has no admissible nontrivial kernel modulo translations and scalings, one obtains local rigidity. If a kernel appears, the higher-order terms decide whether genuine nonradial branches exist. This is probably the cleanest analytic framework for either proving local uniqueness or finding counterexamples.

Direction 4: limit laws for the first eigenvalue

The eigenvalue question becomes sharper if one studies triangular arrays of random perturbations around a fixed analytic convex domain:

$$r_N(\theta) = r_0(\theta) + \varepsilon_N \sum_{1 \le |n| \le N} a_{n,N} X_n e^{in\theta}.$$

For the Dirichlet Laplacian, assume $\lambda_1(\Omega_0)$ is simple. Shape calculus formally gives

$$\lambda_1(\Omega_N) = \lambda_1(\Omega_0) + \varepsilon_N L(f_N) + \varepsilon_N^2 Q(f_N) + o(\varepsilon_N^2),$$

where f_N is the normal boundary displacement, L is the first shape derivative, and Q is the second variation.

This leads to several possible limiting regimes.

1. If $L(f_N)$ is a sum of many independent Fourier modes and dominates the expansion, then a Gaussian central limit theorem should hold after the appropriate normalization.

2. If Ω_0 is critical under the imposed constraint, for example the disc under fixed area for the Dirichlet first eigenvalue, the linear term vanishes. The leading fluctuation is then quadratic:

$$\lambda_1(\Omega_N) - \lambda_1(\Omega_0) \approx \varepsilon_N^2 Q(f_N).$$

   Depending on the spectrum of Q and the weights $a_{n,N}$, the limit may be a weighted chi-square variable, a second Gaussian chaos, or a Gaussian limit for quadratic forms.

3. If one studies the first nonzero Neumann eigenvalue μ_2, the disc has multiplicity two. Then perturbation theory gives eigenvalue splitting. The first-order limit is not a scalar shape derivative but the eigenvalues of a random 2×2 matrix determined by the boundary perturbation.

4. If high-frequency modes dominate, the problem may enter an oscillatory boundary or homogenization regime. The limit could contain a deterministic spectral shift plus smaller random fluctuations.

A concrete first project is therefore:

> Study random analytic, area-preserving convex perturbations of the disc. Derive the second-order limit law for the Dirichlet λ_1, and separately derive the first-order splitting law for the Neumann eigenvalue μ_2.

This is tractable, uses established shape calculus, and produces genuine limit laws rather than only continuity statements. It also connects naturally to the Schiffer problem because both questions probe how spectral data changes under analytic convex perturbations of the disc.

References to expand later

Useful starting points include Williams' papers on the Pompeiu-Schiffer equivalence and analyticity; Berenstein's inverse spectral work; Garofalo and Segala on univalent functions and the Pompeiu problem; Deng's low-index results for Schiffer in the plane; Canuto's perturbative stability results; Kobayashi's work on perturbations near balls; Mondal's recent centrally symmetric convex result; and standard shape-optimization references such as Henrot and Bucur–Buttazzo.